

Problem

Find $g(t)$ given that

$$G(s) = \frac{20}{(s+1)(s^2+4s+13)}$$

Step-by-step solution

Step 1 of 4

The function $G(s)$ is defined as,

$$G(s) = \frac{20}{(s+1)(s^2+4s+13)}$$

The function $G(s)$ is a proper fraction.

Use partial fraction expansion to break $G(s)$ into simple terms

$$\frac{20}{(s+1)(s^2+4s+13)} = \frac{A}{s+1} + \frac{Bs+C}{s^2+4s+13} \dots\dots (1)$$

Multiplying both sides by $(s+1)(s^2+4s+13)$.

$$20 = A(s^2+4s+13) + (Bs+C)(s+1)$$

$$20 = s^2(A+B) + s(4A+B+C) + 13A+C \dots\dots (2)$$

Step 2 of 4

Substitute -1 for s .

$$\begin{aligned} 20 &= A(1-4+13) + (B(-1)+C)(-1+1) \\ &= A(1-4+13) + (B(-1)+C)(0) \\ &= A(10) \\ A &= 2 \end{aligned}$$

Compare constant terms in equation (2),

$$20 = 13A + C$$

Substitute 2 for A in the equation,

$$\begin{aligned} 20 &= 13(2) + C \\ C &= -6 \end{aligned}$$

Compare s^2 terms in equation (2),

$$\begin{aligned} A + B &= 0 \\ B &= -A \\ &= -2 \end{aligned}$$

Substitute the values of A, B, C in equation (1),

$$\begin{aligned} \frac{20}{(s+1)(s^2+4s+13)} &= \frac{2}{s+1} + \frac{-2s-6}{s^2+4s+13} \\ &= \frac{2}{s+1} - 2\left(\frac{s+3}{s^2+4s+13}\right) \\ &= \frac{2}{s+1} - 2\left(\frac{s+2+1}{(s+2)^2+3^2}\right) \\ &= \frac{2}{s+1} - 2\left(\frac{s+2}{(s+2)^2+3^2}\right) - 2\left(\frac{1}{(s+2)^2+3^2}\right) \end{aligned}$$

Step 3 of 4

The function $G(s)$ becomes as,

$$G(s) = \frac{2}{s+1} - 2\left(\frac{s+2}{(s+2)^2+3^2}\right) - 2\left(\frac{1}{(s+2)^2+3^2}\right)$$

Apply inverse Laplace transforms to the function $G(s)$.

$$L^{-1}(G(s)) = L^{-1}\left[\frac{2}{s+1} - 2\left(\frac{s+2}{(s+2)^2+3^2}\right) - 2\left(\frac{1}{(s+2)^2+3^2}\right)\right]$$

Use linearity property of inverse Laplace transforms to the equation.

$$L^{-1}\left\{\frac{1}{s+a}\right\} = e^{-at}u(t)$$

$$\begin{aligned} L^{-1}(G(s)) &= L^{-1}\frac{2}{s+1} - L^{-1}\left[2\left(\frac{s+2}{(s+2)^2+3^2}\right)\right] - L^{-1}\left[2\left(\frac{3}{(s+2)^2+3^2}\right)\right] \\ &= 2L^{-1}\frac{1}{s+1} - 2L^{-1}\left(\frac{s+2}{(s+2)^2+3^2}\right) - \frac{2}{3}L^{-1}\left(\frac{3}{(s+2)^2+3^2}\right) \end{aligned}$$

Step 4 of 4

Consider the following inverse Laplace transforms,

$$L^{-1}\left(\frac{1}{s+1}\right) = e^{-t}u(t)$$

$$L^{-1}\left(\frac{s+2}{(s+2)^2+3^2}\right) = e^{-2t} \cos 3tu(t)$$

$$L^{-1}\left(\frac{3}{(s+2)^2+3^2}\right) = e^{-2t} \sin 3tu(t)$$

Use the above results in calculation of inverse Laplace transform of $G(s)$.

$$\begin{aligned} L^{-1}(G(s)) &= 2e^{-t}u(t) - 2e^{-2t} \cos(3t)u(t) - 0.667e^{-2t} \sin(3t)u(t) \\ &= \left[2e^{-t} - 2e^{-2t} \cos(3t) - 0.667e^{-2t} \sin(3t)\right]u(t) \end{aligned}$$

Therefore, the inverse Laplace transform of the function $G(s)$ is

$$\boxed{\left(2e^{-t} - 2e^{-2t} \cos 3t - 0.667e^{-2t} \sin 3t\right)u(t)}.$$